

Omar Allam

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Education

Queen's University Bachelor of Applied Science, Electrical Engineering

Expected May 2028

Dean's Scholar Standing

Relevant Coursework: Computer Architecture, Digital Systems, Control Systems, Machine Learning, Signals & Systems

Experience

Qualcomm

Markham, ON

Data Center CPU – Telemetry Firmware Intern

May 2026 – May 2027

- Architected and implemented in **C/Zephyr RTOS** a low-latency shared-memory telemetry system enabling communication among 50+ processors, with interrupt-driven transport designed for lightweight CPU data collection.
- Designed a race-safe **producer-consumer protocol** using DMA, Zephyr threads/work queues, synchronization, cache invalidation and flushing to maintain coherent shared-memory communication across cached processors.
- Owned end-to-end investigation of a **DCPerf video-transcode crash**, using PMU, AMU, Linux perf, and Kratos to profile CPU, memory, NoC, and power behavior and isolate microarchitectural and system-level failure signals.
- Analyzed **IPC, memory bandwidth, front-end/back-end stalls, bus accesses, snoop traffic, and core/cluster states**; expanded root-cause analysis into NoC architecture after evidence indicated an interconnect-related freeze.
- Characterized microsecond-scale power behavior with **Kratos**, correlating OC/UC transitions and voltage/current waveforms with workload execution, CPU states, and failure behavior during targeted system-level debug.
- Drove **software and system architecture** for Qualcomm's data center CPU telemetry implementation, spanning protocol design, concurrency, cache coherence, firmware transport, data collection, and system validation.

Aptera Motors

San Diego, CA

Embedded Systems & Firmware Intern

Jun 2025 – Sep 2025

- Developed embedded **C firmware** for real-time control and telemetry of vehicle electromechanical systems.
- Implemented low-level ADC, GPIO, PWM, CAN, SPI, UART interfaces; debugged hardware-software signal behavior.
- Performed hardware bring-up with oscilloscopes and logic analyzers; resolved timing and integration failures
- Built Python tools for automated testing, telemetry/log analysis, regression detection and hardware debugging.

RTL & Computer Architecture Projects

5-Stage Pipelined MIPS Processor

SystemVerilog RTL

- Designed a **5-stage pipelined MIPS CPU** in SystemVerilog implementing IF, ID, EX, MEM, and WB stages.
- Developed synthesizable **RTL datapath and control logic** for the ALU, register file, PC, pipeline registers, instruction decoder, and memory interface.
- Implemented **branch prediction and pipeline control**; analyzed data, control, and structural pipeline hazards.
- Developed a **SystemVerilog testbench** to exercise instruction sequences and verify control signals, register writes, memory operations, and pipeline behavior.
- Performed **cycle-accurate RTL simulation** to trace instruction flow and diagnose incorrect control and datapath behavior.
- Performed **critical-path analysis** to identify timing-limiting combinational paths and evaluate processor timing.

aQuatonomous Autonomous Surface Vessel

Kingston, ON

Co-Captain & Embedded Systems Lead

May 2025 – Present

- Led embedded and electrical architecture for an autonomous surface vehicle integrating **Jetson, Pixhawk, CAN, ROS2, sensors, actuators, and distributed power systems**.
- Integrated embedded compute, communication buses, motor control, sensors and power electronics across systems.
- Designed fault-handling and system-health logic; validated hardware via bench test, telemetry and field tests.

Technical Skills

RTL Microarchitecture: SystemVerilog, Verilog, FPGA, pipelining, branch prediction, FSMs, hazard analysis, critical-path analysis, RTL simulation

CPU Performance Architecture: PMU, AMU, Linux perf, NoC, cache coherence, shared mem, performance profiling

Programming Systems: C, C++, Python, Zephyr RTOS, bare-metal, Linux, Git/GitHub

Hardware Tools: CAN, SPI, UART, I2C, oscilloscopes, logic analyzers, Kratos, KiCad, LTspice, MATLAB/Simulink